

From Bioeconomy Promise to Biodiversity Ceiling: The Role of Forest Biomass in Finnish Decarbonisation[‡]

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Abstract

Forest biomass supplies around 25% of Finland’s primary energy, yet most whole-energy system models treat it as a homogeneous, unconstrained resource – an assumption that ignores real biodiversity ceilings, carbon-sink constraints, and physical disturbance risks. This paper presents the application of EnergyScope TD, an open-source linear-programming whole-energy system model, to Finland, and uses it to examine how contrasting forest-management choices propagate into 2035 decarbonisation pathways. Three contributions are made: a model validation against the 2017 observed energy balance (primary-energy aggregates, electricity mix, and net CO₂ emissions reproduced within approximately 5% on most indicators); the translation of three forest-management narratives – intensive (S1), balanced (S2), and conservation-first (S3) – into differentiated stepwise biomass supply curves; and the analysis of a completed 2035 scenario matrix crossing these narratives with GHG targets. Under the –95% target, the conservation scenario saturates its 40 TWh domestic ceiling and becomes the highest-cost configuration, as the optimiser must substitute or accept expensive imports once the biodiversity constraint eliminates all intermediate domestic steps. Forest-biomass governance choices therefore materially shape both the feasibility and cost of deep decarbonisation. This work is a deliberate first step toward a broader physical-risk research agenda in which climate-driven disturbance shocks – bark beetle outbreaks, windstorms, compound events – will be layered onto these management baselines.

Keywords: Finland; EnergyScope TD; forest biomass; decarbonisation; energy system optimisation.

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1 Introduction

Finland’s pathway to deep decarbonisation sits at the intersection of three system constraints: electrifying end uses at scale, integrating high shares of variable renewables with adequate storage and flexibility, and managing a uniquely large forest biomass resource under increasing scrutiny about carbon accounting and biodiversity impacts. Forests cover roughly 75% of Finland’s land area, and woody biomass already supplies around 25% of national primary energy – largely through industrial residues, logging by-products, and combustion in district heat and combined heat-and-power (CHP) plants – making the forest sector structurally embedded in both the energy system and the wider bioeconomy (Blattert et al., 2022). In this context, energy planning tools are not only asked to identify least-cost technology mixes, but also to make explicit cross-sector trade-offs between electricity, heat and mobility, and the role of renewable fuels and carbon capture options where direct electrification is difficult.

A common thread in European decarbonisation narratives is a strong reliance on forest biomass as a flexible, dispatchable, and storable renewable resource able to bridge the gap between electrification and residual uncompressible CO₂ emissions. Finland’s own Bioeconomy Strategy exemplifies this bet: it frames the forest as a cornerstone of both climate policy and industrial competitiveness by setting ambitious targets for roundwood mobilisation and high-value bio-based products. Yet this narrative sits in tension with two other imperatives that pull in opposite directions. First, biodiversity safeguards impose a real ceiling on what can actually be harvested: Mönkkönen et al. (2024) show that current timber harvesting in southern Finland already approaches the maximum economically sustainable level (roughly 96% of the mean annual increment), while an ecologically sustainable safe operating space – one that maintains favourable habitat conditions for boreal forest species under the IUCN Red List of Habitats framework – would require reducing harvest to approximately 60% of that ceiling. Second, physical risks amplified by climate change are making the supply itself increasingly uncertain: warming and drying trends are increasing the frequency and severity of windstorms, bark beetle outbreaks, and wildfires, all of which can alter the volume, quality, and spatial distribution of available biomass in ways that static supply assumptions cannot capture.

This paper develops a national application of EnergyScope TD to the Finnish energy system. EnergyScope TD is an open-source, linear programming, whole-energy system model that optimises both investment and operation for a target year, with an hourly representation through a typical-days reduction. This makes it suitable for scenarios with high penetration of intermittent renewables and storage (Limpens et al., 2019, 2024). By construction, it provides an open-source and computationally efficient framework for exploring alternative transition constraints.

Methodologically, the first contribution is to build and document a Finland-specific dataset and calibration for a 2035 snapshot analysis, consistent with EnergyScope TD requirements (end-use demands, technology portfolio, resource potentials, and time series for key variables). The second contribution is model verification through historical reproduction. Following the approach used for Switzerland and Belgium, we constrain the model with observed end-use demands and technology shares for a past reference year and compare model outputs to reported energy balances and emissions, as a practical test of the model’s ability to represent a national system (Moret et al., 2017; Thiran, 2025; Limpens, 2021). The third contribution is the construction of differentiated biomass supply curves for Finland under three contrasting forest-management narratives: intensive (bioeconomy-aligned), balanced, and conservation (biodiversity-first). Following the stepwise methodology of Colla et al. (2022), each narrative is translated into a set of biomass availability steps differentiated by feedstock origin, quality, and marginal cost. The purpose is threefold: to make explicit how competing societal choices about forest management propagate into the energy system; to quantify the cost of biodiversity safeguards in terms of system cost and decarbonisation difficulty; and to build a modelling foundation that can, in subsequent work, absorb disturbance-driven supply shocks as perturbations on top of these management baselines.

The overarching question motivating this research agenda is: how do physical constraints on boreal forests – whether structural, from management regimes, or episodic, from natural dis-

turbances – affect the availability and cost of woody biomass, and through that, the economic competitiveness and decarbonisation trajectory of the Finnish energy system? Answering this question in full requires coupling forest dynamics and disturbance models with an energy system optimisation model. The present paper is a deliberate first step: it establishes the management-scenario baselines by studying how different forest-governance choices translate into the energy mix under a 2035 horizon. Subsequent work will layer disturbance shocks – bark beetle outbreaks, windstorms, and compound events – onto these baselines to study how physically induced wood price shocks and supply disruptions propagate through the energy sector. This question is particularly relevant because many whole-energy system models wrongly treat biomass as carbon neutral and with unlimited availability, which can strongly distort optimal outcomes in biomass-rich countries such as Finland.

The specific research questions addressed in this paper are:

1. **Model credibility:** To what extent can a constrained EnergyScope TD setup reproduce Finland’s observed energy balances and combustion emissions for a past reference year (within acceptable error margins), providing confidence for forward-looking scenario analysis?
2. **Biomass supply and management narratives:** How do contrasting forest-management scenarios – from intensive bioeconomy-aligned harvesting to biodiversity-constrained conservation – translate into differentiated supply curves, and how much forest biomass is allocated across heat, power, industry and transport under each narrative?
3. **Constraint interaction:** How do tighter forest-biomass ceilings interact with a -95% GHG target, or with a nuclear phase-out and what cost penalty emerges when forest biomass can no longer act as a flexible adjustment margin?

The remainder of the paper is organised as follows. Section 2 reviews the closest literature and the broader modelling context. Section 3 describes the Finnish EnergyScope TD implementation, the main data inputs, and the qualification of the 2035 forest-biomass scenarios and supply curves. Section 4 presents the 2017 validation, the 2035 scenario results, and the disturbance-shock extension that follows from the present modelling boundary. Section 5 concludes.

2 Literature and context

2.1 Forest Management Scenarios, Physical Risks, and Remaining Gap

The central challenge this paper engages with is the contested nature of Finland’s forest biomass: an abundant and structurally embedded resource whose future availability depends both environmental conditions and forest management choices. [Blatter et al. \(2022\)](#) show that Finland’s Bioeconomy Strategy, National Forest Strategy, and Biodiversity Strategy imply markedly different management mixes and ecosystem-service outcomes, revealing that forest biomass is not a neutral physical input but the product of conflicting policy priorities. [Mönkkönen et al. \(2024\)](#) sharpen this point by arguing that current harvesting in southern Finland already approaches the maximum economically sustainable level, while an ecologically sustainable safe operating space – one that maintains favourable habitat conditions under the IUCN Red List of Habitats framework – would require a drastic reduction to roughly 60% of that ceiling, with a fundamentally different balance between protection and intensive management. In a country this richly endowed in forest resources, the question is therefore not simply how much wood exists, but which forest system society is willing to legitimate.

At the European scale, regulatory pressure amplifies these domestic tensions. The EU Biodiversity Strategy for 2030 (EUBDS), which mandates strict protection of 10% of EU forests and closer-to-nature management for an additional 20%, has been modelled to reduce EU roundwood

production by up to 48% under intensive protection scenarios, with 50–73% of this deficit compensated by increased imports from non-EU countries with weaker forest governance – shifting rather than reducing global pressure on biodiversity (Fischer et al., 2024; Schier et al., 2022; Kallio and Rannestad, 2025; Rosa et al., 2023). Policy safeguards such as deforestation-free supply chain regulations offer only partial mitigation unless sustainability criteria are enforced globally (Fischer et al., 2024; Zamani and Pelikan, 2026). For Finland – one of Europe’s largest roundwood producers – these EU-level dynamics add a regulatory dimension to the domestic trade-off but do not resolve it.

A distinct and growing constraint on biomass availability comes from physical disturbances. There is now strong evidence linking climate change to an increasing frequency and severity of forest disturbances – storms, pest outbreaks, fires, and droughts – across the boreal and temperate zones (Anderegg et al., 2022; Lecina-Diaz et al., 2021). For Finland specifically, Venäläinen et al. (2020) identify wind as the dominant abiotic damage agent in Finnish boreal forests and show that the shortening of the soil frost period – which normally anchors tree roots – is increasing windthrow susceptibility across the most productive southern stands. A key non-linearity compounds this risk: wind-damaged areas generate large volumes of weakened or dead trees that become ideal breeding substrate for bark beetles, directly coupling abiotic and biotic disturbances in potentially multi-year cascades. These events exhibit fat-tailed uncertainty distributions that make both occurrence and magnitude difficult to project reliably, and their economic consequences are asymmetric: disturbances can trigger a short-term surge in timber supply through salvage logging, but typically produce long-term declines in forest productivity as damaged cohorts fall out of productive rotations for decades.

The Czech bark beetle crisis provides the clearest recent illustration of this dynamic. Beginning around 2015, the Czech Republic experienced a severe outbreak of the spruce bark beetle (*Ips typographus*), intensified by elevated temperatures and extended drought periods (Hlásny et al., 2021), with a documented transition from wind- to drought-driven outbreak dynamics. By 2019, salvage logging constituted 92% of the nation’s total timber harvest, amounting to 25.5 million cubic metres (Šimůnek et al., 2024). The abrupt increase in supply led to market oversaturation and a consequent collapse in timber prices; despite the higher physical volume, the forestry sector incurred economic losses estimated at EUR 1.12 billion (Toth et al., 2020). The widespread destruction of mature spruce stands also raises long-term concerns for forest regeneration, carbon sequestration, and future timber availability, potentially undermining the role of forests in climate mitigation.

Translating such disturbances into their energy-system implications is rare in the literature. Rečka et al. (2023) represent the closest methodological precedent: using a spatially enriched partial equilibrium framework for energy market, TIMES, disaggregated to NUTS3 regions in Czechia, they incorporate four bark beetle infestation scenarios into forward-looking decarbonisation pathways to 2050. Their headline finding is that the infestation does not durably alter aggregate decarbonisation trajectories – the effect on the renewable energy share remains below 1.5 percentage points and dissipates after 2030, because carbon-sink dynamics (captured through LULUCF accounting) dominate over fluctuations in biomass energy availability. However, tighter ecological constraints on forest biomass increase overall system costs and worsen the prospects of meeting renewable energy targets, while restricting biomass to residential uses forces other sectors to substitute with natural gas. This asymmetry – a modest aggregate disturbance effect but a real and persistent cost penalty from ecological constraints – suggests that the design of the biomass supply envelope matters at least as much as the occurrence of a disturbance shock itself, and motivates modelling frameworks that can represent both in a whole-energy system context.

The gap is that these strands remain weakly connected. Energy-system studies close to our question generally treat biomass availability as an exogenous scenario input; forest-policy studies stop before whole-energy-system reallocation; and disturbance studies rarely enter national optimisation models as explicit biomass supply and price shocks. The Rečka et al. (2023)’s study is a notable exception, but it addresses disturbance dynamics in a non-forest-rich country rather than the political construction of supply ceilings in a country where biomass is struc-

Model	Multi-sector	Open-source	Optimisation	Comp. time
Calliope	✓	✓	✓	mins
COMPOSE	✓	✗	✓	-
DESSTinEE	✓	✓	✗	s
DIETER	✗	✓	✓	mins
EnergyPLAN	✓	✗	operation only	s
MARKAL/TIMES	✓	✗	investment only	5-35 min
oemof	✓	✓	✓	mins
OSeMOSYS	✓	✓	✓	mins
PyPSA	✓	✓	✓	mins
STREAM	✓	✗	✗	-
Switch	✗	✓	✓	10-60 min
URBS	✗	✓	✓	20 min
EnergyScope TD	✓	✓	✓	~1 min

Table 1: Indicative comparison of large-scale energy planning models along four criteria commonly used in the energy optimisation model literature: multi-sector scope, open-source status, optimisation of both design and operation, and order-of-magnitude runtime per scenario. Adapted from the comparison reported in [Limpens et al. \(2019\)](#).

turally central. This paper addresses the first part of that gap by translating contested Finnish forest-management narratives into explicit biomass supply ladders within EnergyScope TD and by testing their implications for a completed 2035 forest $\times$ GHG scenario matrix. Disturbance shocks are not yet implemented, but the evidence above provides the rationale for treating them as the next extension of the framework rather than as a side issue.

2.2 Energy system modelling context

EnergyScope TD is a whole-energy system optimisation model designed for scenario analysis of regional or national energy transitions with high temporal resolution while keeping computational times compatible with extensive sensitivity and uncertainty analyses. In contrast with power-only optimisation tools, EnergyScope TD represents power, heating, mobility and, non-energy demand such as ammonia within a single, energy balance framework. A central design choice is the use of a limited set of representative "typical days" (TDs) to approximate intra-annual variability, combined with a reconstruction mechanism that preserves inter-day storage and capacity dynamics.

This modelling choice responds to a broader pattern identified in the deep decarbonisation literature. Based on a review of European decarbonisation scenario studies, [Borasio and Moret \(2022\)](#) highlight both the diversity of modelling practices and a recurrent trade-off between scope and tractability: many studies rely either on top-down approaches or on tools primarily centered on the power sector, while fewer frameworks represent sector coupling across power, heat and mobility in a consistent optimisation setting. This motivates the use of whole-energy system models that can capture cross-sectoral substitutions and flexibility needs while remaining sufficiently lightweight for systematic scenario exploration.

A key motivation for EnergyScope TD is therefore that combining (i) multi-sector coverage, (ii) open-source availability, (iii) endogenous system design and operation, and (iv) short run-times is uncommon in large-scale energy planning practice. Table 1 summarises this positioning using the comparative criteria reported in [Limpens et al. \(2019\)](#).

EnergyScope TD is part of a broader open modelling family. Historically, EnergyScope was first developed with coarser temporal aggregation and later extended toward hourly resolution

with TDs and seasonality in EnergyScope TD, and further toward multi-regional representations in EnergyScope Multi-Cell (Hernandez et al., 2021; Thiran, 2025; Borasio and Moret, 2022).

A common critique of EnergyScope TD, also emphasised in reviews of deep decarbonisation modelling, concerns spatial representation. Single-node national models may abstract from intra-country congestion, regional renewable heterogeneity, and the location of flexible demand, which can matter when electrification and variable renewables expand rapidly (Borasio and Moret, 2022; Limpens et al., 2024). In the Finnish case, however, a first-order single-region representation is arguably more defensible than in countries with multiple dense load centers, because population, most energy demand, and the bulk of industrial activity are concentrated in the South, where the main transmission backbone and urban heat loads are also located. This motivates using EnergyScope TD for the present Finnish application.

In climate-mitigation narratives, biomass is often framed as a renewable and flexible resource, yet estimates of its future contribution remain highly uncertain because they depend on the constraint set adopted, including sustainability rules, technical feasibility, land and biodiversity limits, and economic conditions. Under sustainability-oriented assumptions, Beringer et al. (2011) report a wide global range for bioenergy in 2050 (126–216 EJ), broadly consistent with other assessments that apply different constraint sets and therefore obtain markedly different potentials (Haberl et al., 2010; Schueler et al., 2013; Searle and Malins, 2015; IEA, 2023). At present, biomass already constitutes a substantial fraction of renewable energy supply and is used predominantly for heat (IEA, 2020, 2023). Looking forward, deep decarbonisation implies defossilising multiple sectors simultaneously, which can increase competition for lignocellulosic resources as they are called upon to provide not only heat and power but also transport fuels and carbon-based feedstocks for industry and chemicals (Daiglou et al., 2015). Hence, the key transition question is not only *how much* biomass can be mobilised, but also *where* it should be allocated across final needs once economy-wide greenhouse-gas constraints bind (Colla et al., 2022).

A central theme in the biomass literature is that pathway comparisons should be made at the level of delivered services rather than solely at the level of produced fuels. Substitution logics distinguish fuel-level substitution from energy-service substitution, where the avoided emissions depend jointly on conversion efficiencies and on the demand-side technologies that ultimately use the energy carrier (Wahlund et al., 2004). In this view, conversion losses in biofuel production may be compensated if the carrier enables substantially more efficient end-use devices (Steubing et al., 2012). This mechanism becomes visible in whole-system optimisation: Codina Gironès et al. (2018) compare woody-biomass conversion routes inside a national energy-system optimisation and show that biofuel pathways can outperform direct biomass combustion in avoided CO₂ only when accompanied by a broader shift toward efficient end-use options (for instance heat pumps or battery electric vehicles).

Taken together, these insights motivate the use of whole-energy system optimisation frameworks that decide endogenously both (i) how much biomass to use and (ii) how to distribute it across heat, electricity, mobility fuels, and material uses, conditional on the rest of the energy system and on emissions constraints. In that spirit, Colla et al. (2022) implement a detailed lignocellulosic biomass representation in EnergyScope TD in Belgium and analyse how optimal allocations shift as GHG constraints tighten, providing a documented modelling base that can be transferred to new country contexts.

3 Model description

This section presents the implemented Finnish EnergyScope TD setup, including the two-step temporal aggregation and optimisation workflow, the main national input data used to calibrate demand and supply, and the 2035 forest-biomass scenario qualification translated into supply curves.

3.1 Two-step methodology and data flow

EnergyScope TD is typically implemented as a two-step workflow: (i) a temporal aggregation step that selects a small set of TDs (and their weights and sequence), and (ii) a linear programming (LP) optimisation step that determines the least-cost (or otherwise constrained) energy system configuration for a target year under the TD representation (Limpens, 2021).

The workflow therefore combines a temporal aggregation step and a system optimisation step. In practice, Step 1 selects representative days and reconstructs a chronological year, while Step 2 solves the least-cost energy-system configuration on that reduced time basis, following the general EnergyScope TD logic documented by Limpens et al. (2019).

3.2 Step 1: Selecting typical days

In the EnergyScope framework, Step 1 is dedicated to the selection of a subset of representative days, known as typical days (TDs), to overcome the "curse of dimensionality" and reduce the computational time of the energy system optimisation from over 19 hours to approximately one minute.

3.2.1 Inputs and preprocessing

Step 1 starts from hourly time series intended to represent the main drivers of intra-annual variability in supply and demand. In the EnergyScope TD applications discussed by Limpens (2021), the TD selection is built from a small set of demand and weather series, including electricity demand, heat demand, solar irradiance, and wind (onshore and offshore) profiles (See Table 2). The series are normalised and weighted to ensure that the TD set captures the most decision-relevant variability for system design (for instance, by giving more emphasis to electricity and heating profiles when studying flexibility and storage).

For Finland, we rely first on the pre-compiled hourly profiles distributed with the EnergyScope Multi-Cells database Thiran (2025). All time series are provided on a common historical reference year (2017) to ensure synchronicity across demand drivers and weather-dependent renewable availability. These inputs are used both to construct the typical days and to drive the hourly constraints of the optimisation model.

Demand-side series are expressed as normalised hourly profiles (shares over the year), which makes the typical-day selection focus on temporal patterns rather than absolute annual levels. Weather-dependent renewable series are expressed as capacity factors, preserving intra-day and seasonal variability in PV, wind, and hydro. For Finland, this implies that space heating drives strong winter seasonality, while PV adds a pronounced day-night and summer-winter structure, and wind contributes irregular day-to-day variability.

To document the typical-day construction step, Figures 1–4 display the pre-compiled hourly time series used as inputs in EnergyScope Multi-Cells for Finland (reference year 2017, UTC). Demand drivers are expressed as normalised hourly profiles (shares of annual totals), while renewable availability is expressed as capacity-factor type profiles.

The electricity profile exhibits both intra-day structure (systematic daily cycles) and a marked seasonal component, with higher intensities during the cold season. Low-temperature space-heating demand is strongly seasonal, with a pronounced winter concentration and near-zero levels in summer, reflecting its degree-hour construction. Space-cooling appears as a sparse summer phenomenon, indicating that cooling is a minor driver in the Finnish context and contributes limited discriminatory information outside the warmest weeks. Because all demand series are normalised, the colour intensity should be read as the relative allocation of annual demand across hours rather than absolute load levels.

Table 2: Hourly time series used in EnergyScope Multi-Cells for Finland to build typical days (TDs) and to drive the optimisation model (Thiran, 2025). All profiles are synchronous and based on a common reference year (2017, UTC).

Series (file column)	Meaning in the model	Main data source and processing
ELECTRICITY	Specific electricity hourly share (normalised)	ENTSO-E Transparency load (via OPSD), subtract electric heating and cooling (HRE4-based dispatch), then normalise.
HEAT_LOW_T_SH	Space heating hourly share (normalised)	OPSD country temperature from MERRA-2; compute heating-degree-hours with 14°C threshold; apply 24h moving average; normalise.
SPACE_COOLING	Space cooling hourly share (normalised)	OPSD country temperature from MERRA-2; compute cooling-degree-hours with 20°C threshold; apply 24h moving average; normalise.
MOBILITY_PASSENG.	Passenger mobility hourly share	NHTS-based hourly profile, kept as in EnergyScope TD Belgium.
MOBILITY_FREIGHT	Freight mobility hourly share	In the released dataset this profile is flat over the year (no temporal structure).
PV	PV capacity factor (GW/GWp)	Renewable.ninja country-aggregated profiles (via OPSD); used for rooftop and utility PV.
WIND_ONSHORE	Onshore wind capacity factor	Renewable.ninja country-aggregated profiles (via OPSD).
WIND_OFFSHORE	Offshore wind capacity factor	Renewable.ninja profiles (via OPSD), with reconstruction or proxy rules for missing historical series.
SOLAR	Solar thermal proxy capacity factor	Derived from GHI using OPSD weather data computed from MERRA-2.
CSP	CSP thermal output proxy	DNI from PVGIS (via pvlib) and thermal output computed with Oemof thermal; representative plant location.
HYDRO_RIVER	Run-of-river capacity factor	Daily run-of-river generation (JRC-EFAS, completed by ENTSOE-PECD), interpolated to hourly and normalised.
HYDRO_DAM	Hydropower inflow proxy capacity factor	Weekly inflow (JRC-EFAS, completed by ENTSOE-PECD), interpolated to hourly and normalised.
TIDAL	Tidal capacity factor	Literature-based tidal profile, rescaled to match an average capacity factor assumption.

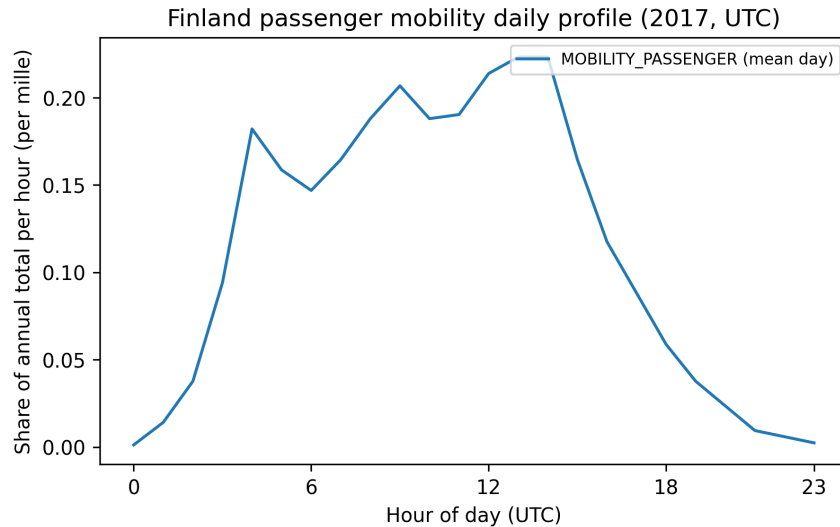


Figure 2: Passenger mobility daily profile used in EnergyScope Multi-Cells (Finland application, 2017, UTC). The profile is adopted from the EnergyScope TD Belgium setup, itself based on a generic household travel survey hourly pattern, and repeated identically for each day of the year (time-zone mapped).

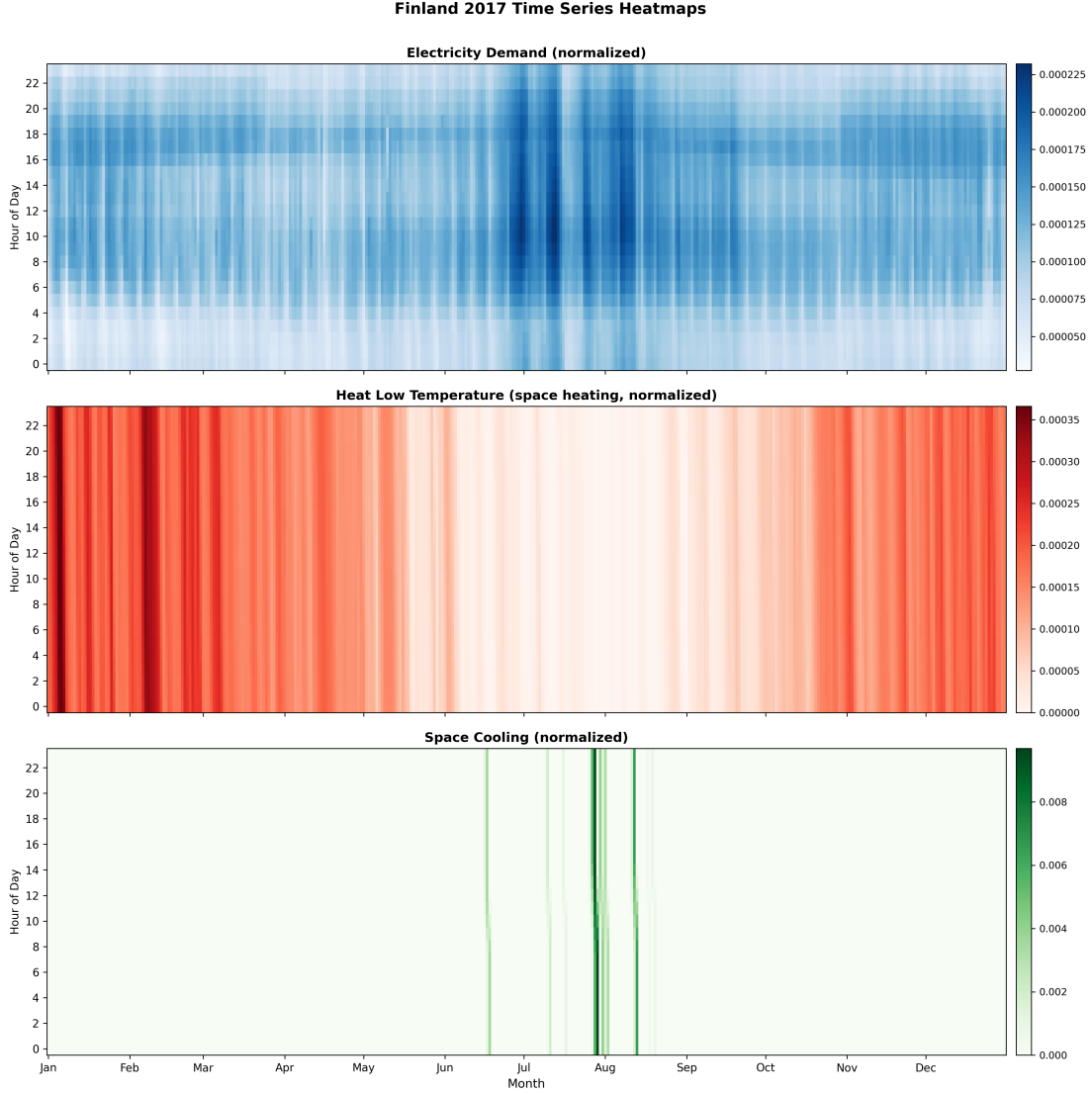


Figure 1: Finland (2017, UTC): demand-driver heatmaps used in the typical-day preprocessing. Each panel shows the distribution of hourly demand over the year (normalised profiles), with months on the x-axis and hour of day on the y-axis.

The passenger mobility profile is characterised by a morning ramp-up and a broad daytime plateau, followed by a steady decline in the evening. In this modelling framework, the curve is exogenous and does not vary across days, so it does not drive the identification of representative days in clustering. Its role is instead to provide within-day structure for mobility demand once typical days have been selected, ensuring consistent diurnal timing of transport services across scenarios.

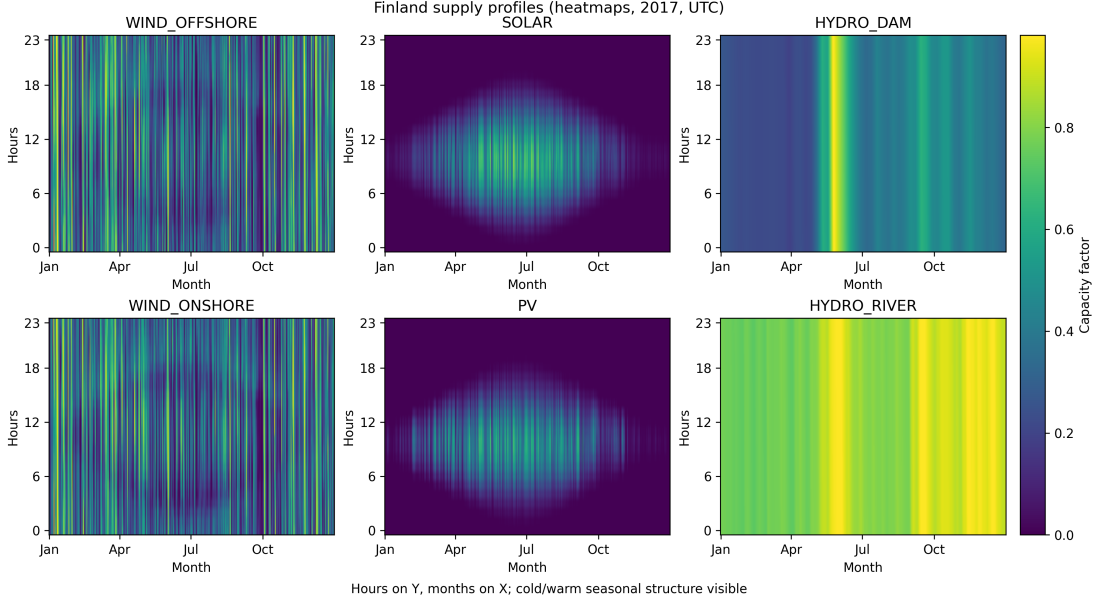


Figure 3: Finland (2017, UTC): renewable availability heatmaps used as TD inputs. Panels report capacity-factor type hourly profiles for wind (onshore and offshore), solar (thermal proxy) and PV, plus hydropower proxies (dam inflow and run-of-river), and tidal (negligible for Finland, so not plotted)

Solar and PV exhibit the expected joint diurnal and seasonal structure: near-zero availability during winter and night-time, with sustained daytime generation during summer months. Wind availability shows substantial hour-to-hour and day-to-day variability across the full year, providing strong discriminatory power for typical-day selection. Hydropower proxies are comparatively smoother, with changes that are more seasonal than diurnal, consistent with inflow-driven dynamics and the aggregation procedure used to derive normalised profiles. Together, these patterns motivate including multiple supply-side drivers in the clustering so that typical days reflect both short-term intermittency (wind) and seasonal solar scarcity.

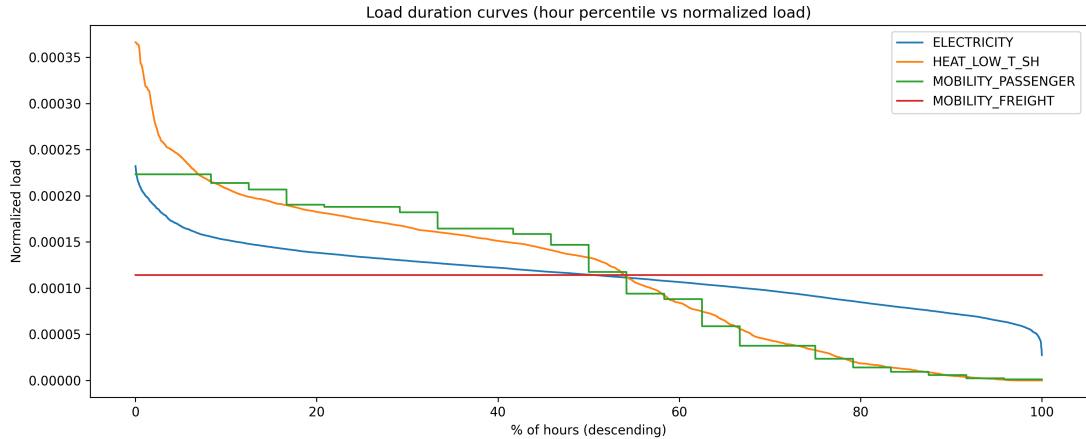


Figure 4: Finland (2017, UTC): load duration curves for normalised demand drivers. Curves show the ordered distribution of hourly values (descending), highlighting peakiness and the weight of extreme hours in annual demand profiles.

The electricity duration curve is relatively smooth, indicating that a large portion of annual

electricity demand is spread across many hours, with limited reliance on extreme peaks. By contrast, the low-temperature space-heating curve displays a steeper tail, consistent with winter-driven peaks that can materially affect capacity sizing and adequacy constraints. Passenger mobility exhibits a stepped structure because the same diurnal pattern is repeated throughout the year, while freight is flat in the compiled dataset and therefore appears as a constant line. These duration curves provide a compact diagnostic for the extent to which typical days must preserve peak and near-peak hours in order to reproduce sizing-relevant extremes.

As an additional diagnostic, Figure 5 reports daily-mean co-variation between selected demand and renewable drivers. While typical-day clustering operates on 24-hour vectors rather than daily averages, these scatter plots clarify cross-variable seasonal structure and motivate retaining multiple drivers in the clustering space.

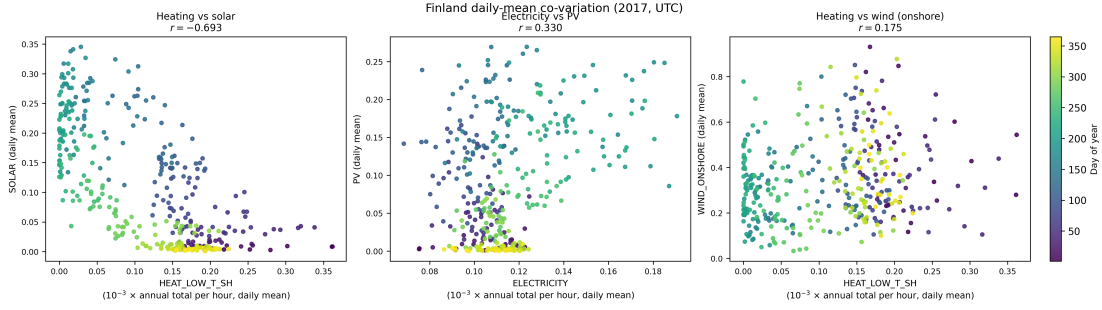


Figure 5: Finland (2017, UTC): daily-mean co-variation between selected TD drivers (colour indicates day of year). Demand variables are shown in per-mille of annual total per hour. Reported r are Pearson correlations across daily means.

Heating and solar availability are strongly anti-correlated ($r = -0.693$), reflecting a seasonal mismatch in which high space-heating needs coincide with low solar resources. Electricity demand and PV availability display a weaker positive association ($r = 0.330$), consistent with both being higher during brighter months while remaining dispersed due to day-to-day weather and non-climatic electricity drivers. Heating and onshore wind show near-independence ($r = 0.175$), indicating that wind intermittency is largely orthogonal to heating seasonality; this supports retaining both heating and wind in the feature set so that typical days capture jointly stressful configurations (e.g., cold days with low wind).

In the typical-day construction, we prioritise the most informative drivers of hourly system constraints, namely electricity demand, low-temperature space-heating demand, space-cooling demand (when present), and the main weather-dependent renewable capacity factors (PV and wind, plus run-of-river hydro when available). By contrast, profiles that are flat or nearly invariant across days (for example freight demand in the compiled dataset) provide little discriminatory power for day selection and are therefore not used as primary clustering attributes.

3.2.2 Clustering and representative-day selection

Let $x_{k,d,h}$ denote the value of time series k at hour h of day d . A clustering procedure partitions the $D = 365$ days into N clusters, each represented by one TD. In the EnergyScope TD literature, several clustering and representative-day selection strategies are compared, including k-medoids and alternative heuristics; their relative performance depends on the trade-off between computational effort and approximation accuracy (??). The output of this stage is: (i) a set of TDs, (ii) a mapping $td(d)$ assigning each original day to a TD, and (iii) weights w_{td} (the number of original days represented by each TD).

3.2.3 Chronological reconstruction and storage coupling

A limitation of TD-based representations is that they can break chronology, which is problematic for storage technologies whose optimal operation depends on sequences of days rather than isolated daily shapes. EnergyScope TD addresses this issue by coupling TD operation with a chronological reconstruction of the year, so that storage state variables evolve over the full horizon while most operational decisions remain defined on the reduced TD set (?Limpens et al., 2019).

The underlying idea can be understood over a one-week sequence. Calendar days are represented as an ordered list of TD labels (for example, TD 1 on Monday, TD 2 on Tuesday and Wednesday, and TD 3 on weekend days). For a given TD, the intra-day charging and discharging power profile is repeated whenever that TD appears in the sequence, but the stored-energy level is not repeated day by day. Instead, the state of charge is tracked continuously across the entire chronology, linking the end-of-day storage level to the start of the next day even when successive days correspond to different TDs. In practice, this coupling allows the model to exploit TDs to capture typical within-day patterns while still representing inter-day and seasonal storage strategies, such as progressively building reserves across sequences of surplus days and drawing them down during extended deficit periods.

3.2.4 Optimising the TD set: error metrics and MILP formulation

EnergyScope TD formalises TD selection using an optimisation-based procedure (MILP) that aims to reduce the approximation error of the reduced time representation. Two complementary error notions are commonly used:

- **Clustering (chronological-shape) error**, measuring how well each day is represented hour-by-hour by its assigned TD.
- **Duration-curve error**, measuring how well the reduced representation reproduces the sorted distribution of hourly values (important for peaks and scarcity periods).

A generic formulation (notation adapted from Limpens et al. (2019)) is:

$$E_{\text{clust}} = \sum_{k \in \mathcal{K}} \alpha_k \sum_{d=1}^D \sum_{h=1}^{24} (x_{k,d,h} - \tilde{x}_{k,td(d),h})^2, \quad (1)$$

$$E_{\text{dc}} = \sum_{k \in \mathcal{K}} \beta_k \sum_{t=1}^{8760} (\text{DC}_k(t) - \widetilde{\text{DC}}_k(t))^2, \quad (2)$$

where $\text{DC}_k(\cdot)$ is the duration curve (sorted hourly values) of series k , and tildes denote the reconstructed values from the TD representation. The MILP chooses the TD set (and assignments) to minimise a weighted combination of these errors subject to selecting exactly N representative days.

Previous EnergyScope TD applications commonly rely on 12 typical days, which has been identified as a robust compromise between temporal accuracy and computational tractability. In the original EnergyScope TD validation, moving from a full 8,760-hour representation to 12 TDs

reduces solve times from multi-hour runs to around a minute, making large scenario ensembles and sensitivity analyses feasible. At the same time, 12 TDs were shown to reproduce the full-year “ground-truth” solution reasonably well for the main key performance indicators used in planning exercises, including primary energy use, technology capacity choices, total system cost, and aggregate greenhouse-gas emissions. Consistent with this established practice, the present Finland application adopts 12 typical days as a baseline; alternative values can be explored later as a robustness check once the dataset calibration and historical reproduction are consolidated.

3.3 Step 2: EnergyScope TD optimisation model

3.3.1 LP problem: resources, layers, and technologies

EnergyScope TD represents the energy system as a network of energy *layers* that must be balanced at each time step. Layers include both resources (e.g. natural gas, biomass, solar) and end-use demand layers (e.g. low-temperature heat, passenger mobility), and technologies connect layers through conversion, storage, or infrastructure. Following the EnergyScope terminology, three technology types are distinguished: (i) end-use type technologies (producing end-use demand layers), (ii) storage technologies (inter-temporal links within a layer), and (iii) infrastructure technologies (supporting conversion, networks, and efficiency).

EnergyScope TD is formulated as an LP problem in which the main design variables are installed capacities $F(j)$ for technologies j , and the main operational variables are hourly flows (resource uses and technology outputs) defined over hours h and TDs td (Limpens, 2021). This structure enables rapid solution times that are compatible with extensive scenario sampling.

The LP solved by EnergyScope TD can be read as a network of balance equations and flow constraints: each coloured *layer* corresponds to an energy-balance equation enforced at every hour (and for each typical day), while each connection corresponds to a non-negative flow variable constrained by technology capacities and conversion coefficients. The solver jointly chooses (i) *design* variables, the installed sizes of assets, and (ii) *operation* variables, the hourly routing of energy through competing pathways to meet end-use demands at minimum system cost. In this network view, storage technologies add inter-temporal coupling by shifting flows across hours, while infrastructure introduces losses and capacity limits, so that the optimum emerges from trading off direct supply, conversion, storage, and cross-sector coupling under resource availability and demand constraints.

3.3.2 End-use demand in energy services (EUD)

Following Moret (2017), EnergyScope specifies exogenous *end-use demand* (EUD) in terms of energy services rather than *final energy consumption* (FEC). EUD refers to the useful service delivered to end users (e.g., low-temperature heat, passenger-kilometres, ton-kilometres), whereas FEC depends on the chosen conversion pathway and its efficiency. Model inputs are annual sectoral EUD values $endUsesYear(eui, s)$, aggregated into a yearly vector $endUsesInput(eui)$. These yearly totals are then allocated to period-specific demands $EndUses(eut, t)$ through transparent distribution and split rules (Fig. 6): electricity is distributed uniformly over periods and corrected for grid losses $Loss(Elec, t)$; high-temperature heat is also spread uniformly; low-temperature heat combines space-heating, shaped by an exogenous profile $\%_{sh}(t)$, with hot-water demand, then splits into district heating and decentralised supply using $\%_{Dhn}$ (including DHN losses $Loss(HeatLowTDhn, t)$); finally, passenger and freight mobility are distributed uniformly and partitioned by modal shares $\%_{Public}$ and $\%_{Rail}$. Consistent with the mapping in Fig. 6, lighting is not represented as a separate end-use and is instead included in the aggregate electricity end-use. The allocation and split parameters (e.g., $\%_{Public}$, $\%_{Rail}$, $\%_{Dhn}$, and the space-heating profile $\%_{sh}(t)$) are country-specific, as they reflect national patterns in modal shares, heating infrastructures, and seasonal demand. In the Finnish application, these parameters therefore need to be calibrated to Finland using national energy and transport statistics to ensure that the mapping from $endUsesInput$ to $EndUses(eut, t)$ matches observed end-use structures (see Table 3).

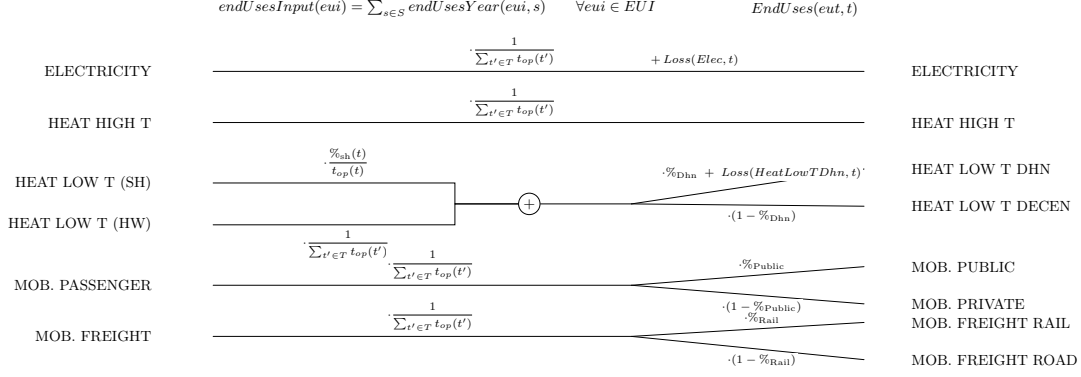


Figure 6: Mapping from yearly end-use inputs $endUsesInput$ to period-specific demands $EndUses(eut, t)$, with lighting aggregated into the electricity end-use. Adapted from ?.

The monthly totals obtained by summing the compiled hourly profiles provide a simple diagnostic of how yearly end-use demand is allocated across the calendar. Because each series is normalised over the year, monthly sums should be read as shares of the annual end-use rather than as absolute energy levels. In the Finnish case, this diagnostic confirms the strong winter concentration of low-temperature space-heating demand, whereas electricity remains comparatively flatter across months and passenger mobility remains close to uniform because its diurnal shape is repeated throughout the year in the compiled dataset. This seasonality check helps identify which end-uses provide the strongest temporal signal that the typical-day selection must preserve.

Table 3: Finland calibration targets for the EUD mapping (baseline year 2020).

Parameter	Symbol	Value	Source
District heating share of low-T heat ^a	$\%_{Dhn}$	0.45	Finnish Energy (2024)
District heating network losses ^b	$Loss(HeatLowTDhn)$	0.085	Finnish Energy (2024)
Inland public passenger mobility share ^c	$\%_{Public}$	0.1608	Statistics Finland (2020), <i>Traffic on highways 2019</i> ; Statistics Finland (2020), <i>Railway statistics 2019</i>
Inland rail freight share ^d	$\%_{Rail}$	0.2761	Statistics Finland (2020), <i>Railway statistics 2019</i> ; Statistics Finland (2020), <i>Goods transport by road 2019</i>
Electricity grid losses ^e	$Loss(Elec)$	0.03	Finnish Energy and Statistics Finland (2022), <i>Energy in Finland 2022</i>

^a Proxy for the DHN split: district heating share of heating energy / space-heating market in Finland (year 2020).

^b Implemented as a constant loss factor; Finnish Energy reports typical DH distribution losses of 8–9% of transmitted energy, hence 0.085 as midpoint.

^c Computed as $(pkm_{bus} + pkm_{rail}) / (pkm_{car} + pkm_{bus} + pkm_{rail})$ using pre-COVID 2019 inland passenger-km: $pkm_{car} = 66.8$ bn and $pkm_{bus} = 7.9$ bn (*Traffic on highways 2019*), and $pkm_{rail} = 4.9$ bn (*Railway statistics 2019*). 2019 is used as a structural proxy to avoid pandemic-related distortions in 2020 modal shares.

^d Computed as $tkm_{rail} / (tkm_{rail} + tkm_{road})$ using pre-COVID 2019 inland freight tonne-km: $tkm_{rail} = 10.3$ bn (*Railway statistics 2019*) and $tkm_{road} \approx 27.0$ bn from domestic road freight performance reported as “close on 27 million tonne-kilometres” in *Goods transport by road 2019* (interpreted as 27,000 million tonne-km).

^e Electricity transmission and distribution losses are set to 0.03 based on the “Losses 3%” entry reported in the electricity consumption breakdown in *Energy in Finland 2022* (compiled from Finnish Energy and Statistics Finland). Used as a constant proxy for baseline year 2020.

3.3.3 Objective function: total annual cost (and emissions accounting)

A standard objective is the minimisation of total annualised system cost, decomposed into annualised investment costs, fixed operation and maintenance costs, and resource operating

costs (Limpens et al., 2019):

$$\min C_{\text{tot}} = \sum_{j \in \mathcal{TECH}} [\tau(j) C_{\text{inv}}(j) + C_{\text{maint}}(j)] + \sum_{i \in \mathcal{RES}} C_{\text{op}}(i), \quad (3)$$

with $\tau(j)$ an annualisation factor that depends on the discount rate and lifetime. In addition, annual system-wide GHG emissions are computed with a life cycle assessment approach, accounting for “cradle-to-grave” emissions from both technologies and energy resources. Climate impacts are reported using global warming potential (GWP) in ktCO₂-eq per year. Total yearly emissions, GWP_{tot} (Eq. (4)), are defined as the sum of (i) the construction and end-of-life impacts of energy conversion technologies, annualised by their lifetime, and (ii) operational impacts from resources, including fuel supply chains (up to combustion) and imported electricity, weighted by the operating period.

Following Limpens et al. (2019):

$$GWP_{\text{tot}} = \sum_{j \in \mathcal{TECH}} \frac{GWP_{\text{constr}}(j)}{\text{lifetime}(j)} + \sum_{i \in \mathcal{RES}} GWP_{\text{op}}(i) \quad (4)$$

3.3.4 Seasonality and long-term storage: coupling typical days

A known limitation of TD approaches is the difficulty to represent inter-day dynamics and seasonal storage. EnergyScope TD implements a reconstruction method in which each day of the year is assigned to a TD through a sequence, while the stored-energy state variable is tracked across all 8760 hours. Operational variables other than stored energy remain defined on the TD basis (Limpens, 2021). This enables modelling of day-to-day and seasonal storage strategies without solving the full hourly year.

Because EnergyScope TD is designed for fast whole-system optimisation, it abstracts from some short-term operational details (e.g. unit commitment or detailed ramping constraints) that are central in dispatch-focused models. The TD approach is therefore best interpreted as a planning-level representation that trades operational detail for tractability, and can be complemented by more detailed operational checks when needed (Thiran, 2025).

3.3.5 Biomass modelling: feedstocks, supply curves, and Finland-specific calibration

Bioenergy potentials are not uniquely defined objects: reported values differ widely because they embed distinct assumptions about sustainability constraints, competing uses (materials, soil functions, circularity), land allocation for energy crops, and waste management pathways. Selecting a given “potential” is therefore a structural modelling choice that shapes the socio-technical system being represented. Colla et al. (2024) illustrate this point by showing that European assessments can imply markedly different 2030 bioenergy ranges for a single country (Belgium: 30–90 TWh across four studies), with divergences driven primarily by (i) how competing uses are accounted for, (ii) land-use rules for energy crops, and (iii) whether waste streams are directed to energy or material recovery. At the European scale, they argue that a “realistic” order of magnitude for EU27+UK in 2030 lies in the lower part of published ranges (about 2000–2500 TWh), noting that forestry feedstocks are already relatively exploited while additional mobilisation would come mainly from agricultural residues, conditional on collection practices and farm structure (Colla et al., 2024). These lessons motivate a transparent, scenario-based calibration strategy for Finland rather than reliance on a single headline potential.

The Belgian application of Colla et al. (2022) presents lignocellulosic biomass through cost supply curves, an origin-by-quality decomposition, and step-specific embedded GHG intensities. Their Belgium case therefore makes three modelling choices explicit: biomass is procured in merit order, different fractions have different marginal costs, and different fractions carry different supply-chain carbon footprints. We retain that logic in the Finnish first stage because it is the

right level of detail for asking where a scarce biomass resource is most valuable once system-wide GHG constraints bind.

The current Finnish implementation does not reproduce the full Belgian origin-by-quality matrix. Instead, it collapses the forest-biomass ladder into four domestic steps and one import backstop. This is a deliberate simplification because the present paper studies forest-management intensity and biodiversity constraints in Finland, not the full geography of European biomass trade. Four simplifications should therefore be stated explicitly:

1. The domestic ladder aggregates the Finnish forest resource into industrial by-products, logging residues, secondary woodchips/sawdust, and direct fuelwood/landscape wood, rather than distinguishing all Belgian origin rings and quality classes separately.
2. Step-specific embedded GHG coefficients are assigned by matching each Finnish step to the most appropriate EU RED II Annex VI supply-chain category as tabulated by [Colla et al. \(2022\)](#) (see Table 4 caption for the per-step mapping). The GHG merit order is not monotone with the cost merit order: mill-gate and industrial wood residues (FI1, FI3) have shorter supply chains than in-forest collected logging residues (FI2), so FI3 carries lower supply-chain GHG (14 tCO₂eq/GWh) than FI2 (22 tCO₂eq/GWh) despite being more costly. Finnish-specific LCA validation would be needed to confirm these values rigorously.
3. The Belgian low-quality share constraint is not re-implemented at this stage. The Finnish first-stage model therefore captures relative cost and relative GHG differences between steps, but not a separate ash-content or quality-mixing constraint.
4. The Finnish scenarios are parameterised to represent forest-policy narratives, not to claim direct measurement of future biomass availability. In particular, S1/BES is stylised and S3/BDS is a conservative audited scenario rather than a raw database value.

Let $r \in \{\text{WOOD_FI1}, \dots, \text{WOOD_FI5}\}$ denote a Finnish biomass step. We introduce decision variables x_r (annual energy procured from step r), with exogenous step capacities $\bar{X}_r$, marginal supply costs c_r , and embedded supply-chain GHG intensities g_r . The first-stage formulation is:

$$0 \leq x_r \leq \bar{X}_r \quad \forall r, \quad (5)$$

$$C_{\text{bio}} = \sum_r c_r x_r, \quad (6)$$

$$\text{GHG}_{\text{bio,supply}} = \sum_r g_r x_r. \quad (7)$$

The optimiser first uses the cheapest domestic fractions and only reaches the high-cost import backstop when cheaper Finnish steps are exhausted.

The neutral reference case S2/NFS is built from the ENSPRESO medium potential for Finland, interpolated to 2035 at the commodity-code level before aggregation into the four domestic steps ([Ruiz et al., 2019](#)). The stylised S1/BES case is then constructed as a +20% scale-up of the S2 ladder. This is not presented as a literal replication of the BES harvest simulated by [Blatter et al. \(2022\)](#); it is a sensitivity case used because the ENSPRESO high scenario for Finland yields implausibly large forest-energy volumes and because the BES narrative increases some residue streams without simply increasing all wood categories proportionally. Importantly, [Blatter et al. \(2022\)](#) find that under its biodiversity non-decline constraint BES achieves only 75% of the maximum possible harvest, with the harvest mix shifting toward more logging biomass and less roundwood relative to NFS. The +20% parameterisation is therefore a deliberate upper-bound sensitivity that follows the BES direction of intensified biomass mobilisation without claiming to replicate the Blatter BES harvest volumes quantitatively. The conservative S3/BDS case is not the original ENSPRESO low value of 74.2 TWh. That initial value was revised downward after a three-way audit combining the biodiversity-constrained forest-management narrative of Blatter

et al., the ecological harvest ceiling proposed by Mönkkönen et al., and a cross-check against recent Finnish national wood-energy statistics (Blatter et al., 2022; Mönkkönen et al., 2024; Natural Resources Institute Finland (Luke), 2024). The revised domestic S3 ceiling is 40 TWh, split across the four domestic steps as 32.0, 6.0, 1.5, and 0.5 TWh. The logging-residue step (WOOD_FI2) also carries a 4 €/MWh cost premium under S3/BDS (26 versus 22 €/MWh in S1/S2). This reflects two BDS-specific cost drivers: lower per-hectare residue yields because the biodiversity strategy requires retaining a fraction of coarse woody debris as deadwood on site (Blatter et al., 2022; Mönkkönen et al., 2025), and longer average extraction distances because conservation set-asides restrict the accessible harvesting area. The resulting premium of $\approx 18\%$ above the S2 base cost is an explicit modelling assumption in the absence of published Finnish per-unit extraction costs for BDS-compliant harvesting; it is flagged as a parametric sensitivity.

Table 4: First-stage Finnish forest-biomass ladder used in the 2035 scenarios. Embedded GHG covers supply-chain only (harvesting, preprocessing, transport); biogenic combustion CO₂ is excluded. GHG values are assigned per step from EU RED II Annex VI supply-chain defaults as tabulated by Colla et al. (2022): **FI2** (logging residues) uses the local forest-residue value (case 3a, ≤ 500 km, $21.6 \text{ tCO}_2\text{eq/GWh} \approx 22$); **FI3** (secondary mill residues) uses the agricultural/short-supply-chain proxy (case 7a, $14.4 \text{ tCO}_2\text{eq/GWh} \approx 14$) as a conservative upper bound for processing-site residues; **FI4** (fuelwood + landscape-care wood) blends the fuelwood component (case 3a, ≈ 22) and the landscape-care component (case 7a, ≈ 14) at an approximate 60/40 ratio, giving $\approx 19 \text{ tCO}_2\text{eq/GWh}$; **FI1** (mill-gate by-products) is set at approximately half the forest-residue baseline ($10 \text{ tCO}_2\text{eq/GWh}$), reflecting the absent in-forest collection and forwarding steps; **FI5** (rest-of-world backstop) uses the long-haul proxy (case 3a, $> 10,000$ km, $39.6 \text{ tCO}_2\text{eq/GWh} \approx 40$). These RED II-derived values are implemented in the model input files used for all scenario runs reported in this paper.

Step	Interpretation and main source mapping	Cost (€/MWh)	Embedded GHG (tCO ₂ eq/GWh)
WOOD_FI1	Industrial by-products from chemical and physical wood processing, primarily black liquor, bark, and sawdust (ENSPRESO MINBIOW00a)	11	10
WOOD_FI2	Logging residues such as branches and tops (ENSPRESO MINBIOFRSR1)	22 in S1/S2; 26 in S3	22
WOOD_FI3	Secondary woodchips and sawdust streams (ENSPRESO MINBIOW00w1 + MINBIOW00w1a)	27	14
WOOD_FI4	Direct fuelwood and landscape-care wood (ENSPRESO MINBIOW00 + MINBIOFRSR1a)	33	19
WOOD_FI5	Rest-of-world biomass import backstop; GHG follows the Colla rest-of-world proxy (39.6 rounded to $40 \text{ tCO}_2\text{eq/GWh}$)	70	40

Table 5: Step capacities (TWh/yr) in the three 2035 forest-management scenarios. S2/NFS values are derived from ENSPRESO medium potentials interpolated to 2035 (Ruiz et al., 2019); S1/BES applies a uniform +20% scale-up to all four S2 domestic steps; S3/BDS capacities are revised from the ENSPRESO low value following a three-way biodiversity-ceiling audit (Blattert et al., 2022; Mönkkönen et al., 2024; Natural Resources Institute Finland (Luke), 2024). Step costs are given in Table 4.

Step	Main feedstock	S1-BES (TWh)	S2-NFS (TWh)	S3-BDS (TWh)
WOOD_FI1	Industrial by-products (black liquor, bark, sawdust)	54.5	45.4	32.0
WOOD_FI2	Logging residues (branches, tops)	45.9	38.2	6.0
WOOD_FI3	Secondary woodchips and sawdust	15.1	12.5	1.5
WOOD_FI4	Fuelwood and landscape-care wood	6.5	5.4	0.5
WOOD_FI5	Baltic/Nordic import backstop		unlimited	
Domestic total		122.0	101.5	40.0

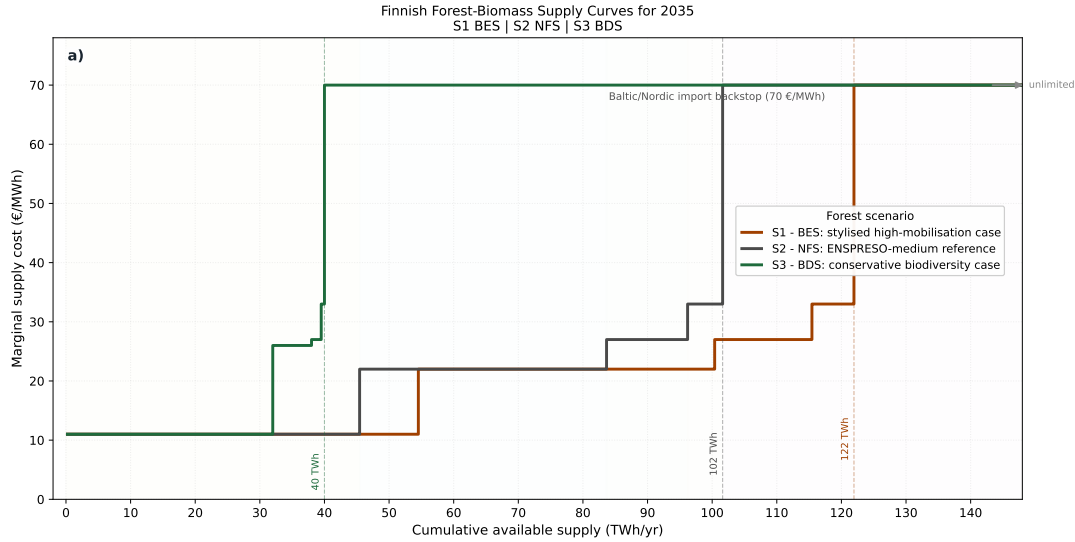


Figure 7: Finnish forest-biomass supply curves used in the 2035 scenario matrix. S1 is a stylised high-biomass sensitivity bound (+20% above the ENSPRESO medium 2035 baseline), S2 the ENSPRESO-medium reference, and S3 a conservative biodiversity-constrained case revised against Blattert et al. (2022), Mönkkönen et al. (2024), and recent Finnish national wood-energy statistics (Natural Resources Institute Finland (Luke), 2025).

Figure 7 presents the three scenario supply curves as classical stepwise cost ladders (cumulative available supply on the horizontal axis, marginal cost on the vertical axis). Several features are worth highlighting. All three curves share the cheapest step (WOOD_FI1: industrial by-products at 11 €/MWh), the least sensitive to forest governance choices because it originates from existing processing plant by-products rather than additional harvesting decisions. Their capacities differ, however, reflecting how much industrial residue each narrative mobilises: 54.5 TWh in S1/BES, 45.4 TWh in S2/NFS, and 32.0 TWh in S3/BDS (Table 5). After FI1, the three scenarios diverge sharply. S3/BDS exhausts its domestic ceiling at 40 TWh after only three further small steps – logging residues (6.0 TWh at 26 €/MWh), secondary woodchips (1.5 TWh at 27 €/MWh), and fuelwood (0.5 TWh at 33 €/MWh) – then immediately jumps to the Baltic/Nordic import backstop at 70 €/MWh: a price discontinuity of more than 40 €/MWh with no intermediate domestic option. By contrast, S2/NFS extends through four

domestic steps up to 101.5 TWh before reaching the backstop, and S1/BES reaches 122.0 TWh. This horizontal compression of the S3 curve illustrates a key modelling insight: biodiversity-constrained forest management does not simply reduce the quantity of cheap biomass while leaving the merit order intact – it eliminates intermediate cost steps entirely and makes the energy system abruptly dependent on expensive imports once the domestic ceiling is reached. The cost consequences of the constraint therefore manifest primarily as a price jump rather than a gradual cost increase, which is why tight forest-biomass ceilings interact strongly with ambitious GHG targets: the optimiser has no moderately priced margin to exploit and must substitute toward other low-carbon options or accept the high-cost import step. A key clarification is that the model’s GHG constraint acts on net combustion CO₂ only; supply-chain emissions are tracked and reported but not separately constrained by the policy budget. The supply-chain correction therefore does not affect optimised system costs, forest-wood quantities, or qualitative technology-substitution results. Second, because the Belgian low-quality share constraint is not yet reintroduced, the Finnish first stage is stronger on transparent supply ranking than on physical feedstock-quality constraints. For the present paper, this keeps the focus on how domestic forest biomass availability changes across management narratives, but future work should strengthen the feedstock-quality treatment.

The retained forward-looking analysis uses a completed 3×3 scenario matrix. The forest dimension consists of S1/BES, S2/NFS, and S3/BDS. The policy dimension consists of an unconstrained case, a -95% GHG case, and a -95% GHG case with full nuclear phase-out. The completed runs show that the unconstrained 2035 system is already strongly decarbonised, whereas the conservative S3 case immediately saturates its 40 TWh ceiling under the -95% target and therefore removes forest biomass as a flexible adjustment margin.

This paper reports the calibrated 2017 benchmark, the static 2035 forest-biomass ladder, and the completed forest-by-GHG scenario matrix. Disturbance shocks (windthrow, bark-beetle outbreaks, wildfire, salvage wood pulses, and dynamic price shocks) remain future work and are treated as the logical continuation of the present study rather than as implemented results.

4 Results

4.1 Finland 2017 Validation

To establish model credibility before turning to forward-looking scenarios, the Finnish implementation of EnergyScope was calibrated against a historical baseline year. The reference run uses 12 typical days and relaxes the CO₂ emission cap to avoid a non-physical forced-CCS artefact in the historical reproduction. The purpose of this benchmark is not to claim predictive validation of the future, but to test whether a constrained historical configuration can reproduce the known structure of the Finnish 2017 energy system with acceptable deviations.

Table 6 summarises the main indicators of the calibrated benchmark against the Finnish 2017 reference values. The main result is that the model reproduces the overall primary-energy balance, electricity mix, and net CO₂ emissions with high fidelity. All major primary-energy aggregates except solar remain within about $\pm 5\%$, electricity imports and CHP generation are also close to the statistical reference, and total net CO₂ emissions are reproduced almost exactly.

This agreement is also visible in Figure 8, which provides the two main graphical validation checks retained in the paper. The left panel shows that the model reproduces the dominant Finnish 2017 primary-energy pillars, especially biomass, oil, gas, coal and nuclear, without requiring large compensating errors across fuels. The right panel shows that the electricity balance is also captured at the right order of magnitude, with the main remaining deviation concentrated in gas-based generation rather than in the low-carbon backbone of nuclear, hydro and wind.

Two residual mismatches should nevertheless be discussed explicitly. First, district-heating production is overestimated by 43.4%. This is not primarily a calibration failure but a structural aggregation artefact: in the present EnergyScope formulation, the district-heating share is applied to the full low-temperature heat pool, including industrial low-temperature demand,

whereas Finnish statistics mainly report residential and service-sector network heat. Second, gas-fired electricity remains 36.7% above the statistical reference because district-heating CHP in the model produces electricity as an unavoidable by-product when gas is routed through CHP technologies. These limitations do not invalidate the historical benchmark, but they define the clearest remaining structural aggregation issues.

Category	Indicator	Actual 2017	Model	Error
Primary energy	Biomass	100.0	96.8	-3.2%
Primary energy	Oil	82.0	80.9	-1.4%
Primary energy	Gas	20.0	20.0	0.0%
Primary energy	Coal + peat	35.0	35.0	0.0%
Primary energy	Nuclear	65.0	65.1	+0.1%
Primary energy	Hydro	15.0	14.6	-2.7%
Primary energy	Wind	5.0	4.8	-4.1%
Electricity	CHP	20.73	20.49	-1.2%
Electricity	Gas power	3.2	4.37	+36.7%
Electricity	Imports	20.43	19.54	-4.4%
Heat	District-heating production	36.5	52.33	+43.4%
Emissions	CO ₂	41.2	41.25	+0.1%

Table 6: Main indicators of the calibrated Finland 2017 benchmark against the statistical reference system. Primary energy and electricity values are in TWh; CO₂ emissions in MtCO₂/y.

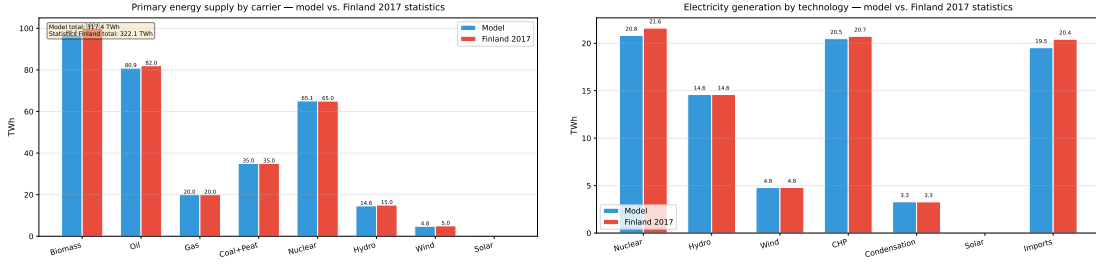


Figure 8: Finland 2017 validation: model versus Statistics Finland. Left: primary energy supply by carrier (TWh). Right: electricity generation by technology (TWh). The main residual deviations are discussed in the text.

For the main text, the retained validation visuals are therefore limited to the primary-energy comparison, the electricity-generation comparison, and a structural Sankey view of the calibrated system. Figure 9 complements Figure 8 by showing that the calibrated benchmark is not only numerically close to Finnish statistics but also structurally coherent as an integrated energy system. In particular, the corrected heat-pump branch makes clear that low-temperature heat is produced from a combination of electricity and ambient heat, while the broader diagram highlights the expected roles of nuclear and hydro in electricity supply, lignocellulosic biomass in heat and CHP, and oil products in mobility and non-energy uses.

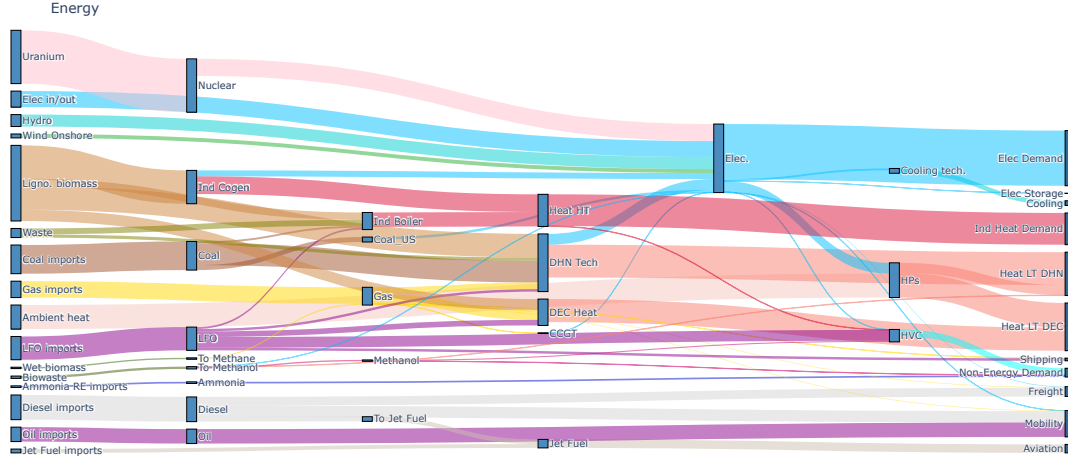


Figure 9: Structural Sankey view of the modelled energy mix for Finland 2017

4.2 Finland 2035 Results Under Forest and Emissions Scenarios

The forward-looking analysis uses the completed 3×3 matrix crossing three forest-biomass supply narratives (S1/BES, S2/NFS, S3/BDS) with three policy configurations (unconstrained, -95% GHG with nuclear, and -95% GHG with full nuclear phase-out). Across the three forest cases, the unconstrained 2035 optimum already yields net CO_2 emissions between about 12.6 and 13.1 MtCO_2/y , corresponding to roughly 68–70% reduction relative to 2017. The policy-relevant question is therefore not whether the unconstrained 2035 system decarbonises, but how the last part of the path toward a -95% target changes when forest biomass becomes scarce and nuclear is removed.

Forest scenario	Policy case	Cost	CO_2 net	Wood used
S1/BES	Unconstrained	22.40 bnEUR/y	13.02 Mt/y	54.5 TWh
S1/BES	-95% GHG (with nuclear)	22.92 bnEUR/y	2.06 Mt/y	74.2 TWh
S1/BES	-95% GHG (nuclear phase-out)	22.94 bnEUR/y	2.06 Mt/y	100.4 TWh
S2/NFS	Unconstrained	22.46 bnEUR/y	12.57 Mt/y	45.4 TWh
S2/NFS	-95% GHG (with nuclear)	23.02 bnEUR/y	2.06 Mt/y	74.2 TWh
S2/NFS	-95% GHG (nuclear phase-out)	23.12 bnEUR/y	2.06 Mt/y	96.2 TWh
S3/BDS	Unconstrained	22.62 bnEUR/y	13.10 Mt/y	32.0 TWh
S3/BDS	-95% GHG (with nuclear)	23.56 bnEUR/y	2.06 Mt/y	40.0 TWh
S3/BDS	-95% GHG (nuclear phase-out)	23.90 bnEUR/y	2.06 Mt/y	40.0 TWh

Table 7: Summary of the retained Finland 2035 forest-biomass scenario matrix. Forest-wood use sums the Finnish forest-biomass ladder resources only.

Table 7 summarises the main results of the retained scenario matrix. Three patterns deserve emphasis. First, the unconstrained cases are relatively close in annualised system cost, but they already differ in biomass use: S3/BDS is more constrained than S1 and S2 even without a climate target. A structural detail worth noting is that in all three unconstrained cases the model saturates exactly the first supply-ladder step (WOOD_FI1, cheap industrial by-products) and draws nothing from the higher-cost domestic steps, indicating that the 2035 Finnish energy demand without a climate target can be met entirely from the cheapest biomass fractions; the supply-curve design therefore matters primarily under the stringent GHG constraint. Second, the -95% case with nuclear fully saturates the conservative S3 ceiling while leaving headroom in S1 and S2. Third, removing nuclear substantially increases biomass demand in S1 and S2 (from 74.2 TWh to 100.4 TWh in S1, and to 96.2 TWh in S2), yet the additional system

cost is surprisingly small: only 0.02 and 0.10 bn EUR/y respectively, because the additional wood is available at the intermediate steps of an unconstrained supply ladder. By contrast, S3 cannot expand further because its 40 TWh domestic ceiling is already binding, making the nuclear phase-out more costly (0.34 bn EUR/y additional). In other words, S3 does not make deep decarbonisation infeasible in the optimiser, but it removes forest biomass as a flexible adjustment margin and therefore becomes the highest-cost configuration, with a sensitivity to nuclear availability roughly three times greater than in S1.

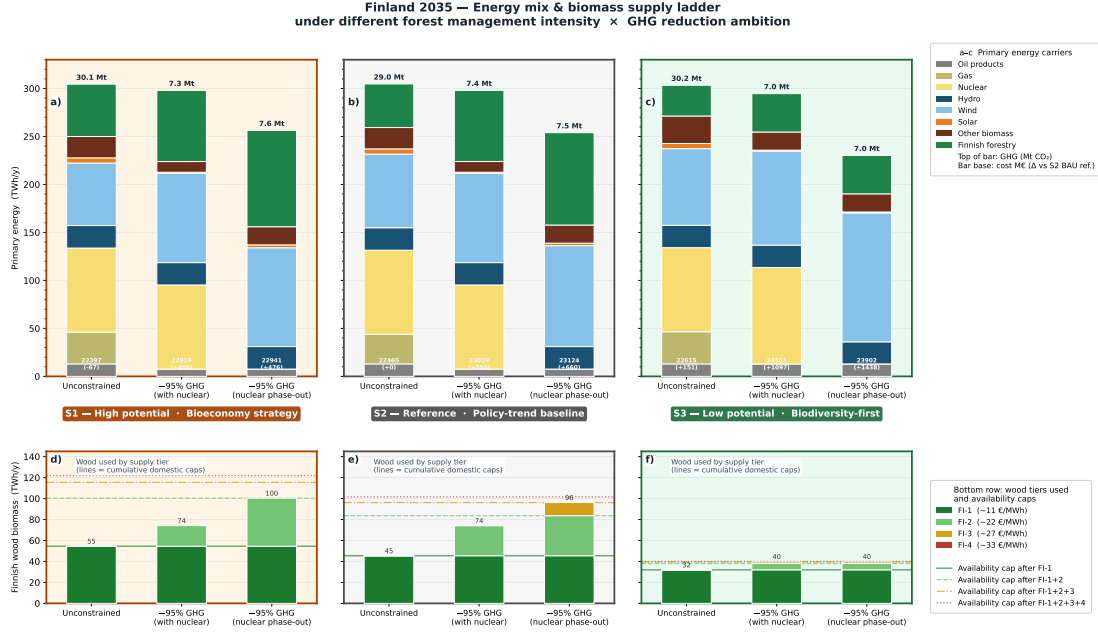


Figure 10: Finland 2035 scenario matrix across three forest-management cases and three GHG configurations. The top row shows the primary-energy mix, while the bottom row shows the use of the Finnish forest-wood supply ladder. The unconstrained cases already reach roughly a 70% reduction versus 2017, while the biodiversity-constrained S3 case is distinguished by early saturation of the domestic wood ceiling under the -95% target.

These system-level shifts are easier to read in Figure 10. The top row shows that the main transition from the unconstrained cases to the -95% cases is not a simple linear reduction in fossil use, but a reconfiguration of the entire primary-energy mix around electricity, nuclear availability, and scarce biomass: fossil imports fall from roughly 46–47 TWh to about 8 TWh across all three forest cases, a reduction of more than 80%, while additional renewables and biomass fill the gap. The bottom row shows that in the unconstrained cases only the cheapest FI1 step (industrial by-products) is mobilised, whereas the -95% cases unlock the next supply tier – FI2 logging residues add 19.6 TWh in S1, 28.7 TWh in S2, and S3 activates all four domestic steps to reach its 40 TWh ceiling. The no-nuclear variants push further up the ladder in S1 (FI2 nearly exhausted at 45.9 TWh) and into the FI3 step in S2 (adding 12.5 TWh). This is the key result of the scenario exercise: biodiversity-constrained forest supply does not prevent deep decarbonisation in the model, but it removes cheap domestic biomass as an adjustment margin precisely when the climate constraint becomes stringent.

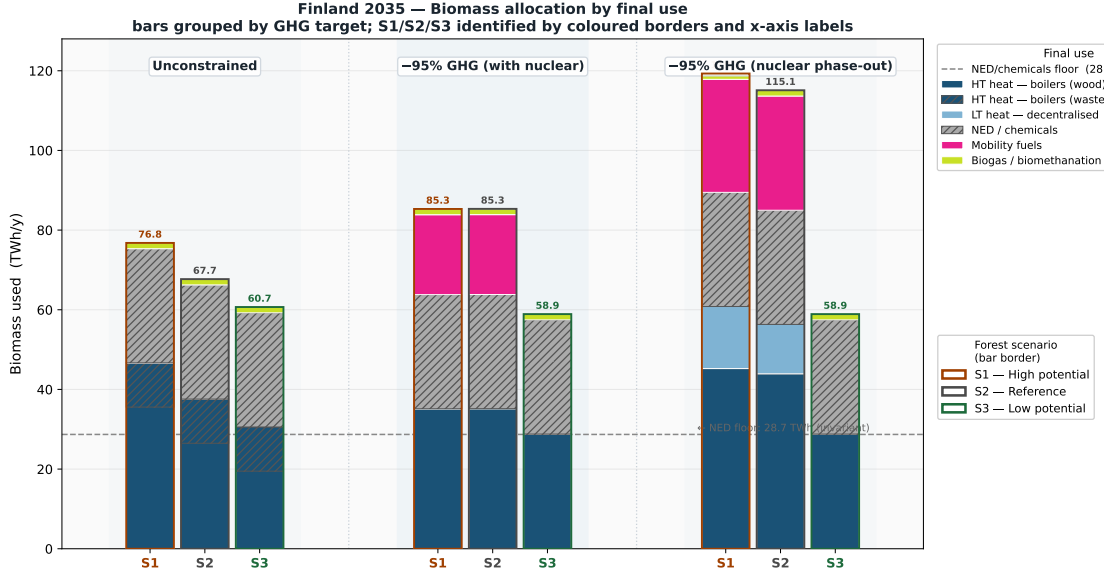


Figure 11: Biomass allocation by final use in Finland 2035. The figure highlights how the no-nuclear cases increase pressure on biomass in S1 and S2, whereas S3/BDS remains capped by its conservative forest-biomass ceiling and therefore relies more heavily on non-biomass adjustments elsewhere in the system.

Figure 11 adds the end-use interpretation behind that aggregate picture. Three patterns stand out.

First, roughly 27.9 TWh is directed to high-value-chemicals production and approximately 0.8 TWh to methanol synthesis across all nine scenarios – a near-constant baseline driven by exogenous non-energy industrial demand that the optimiser cannot reroute. This commitment means that the full S3/BDS ceiling of 40 TWh retains only about 12 TWh for heat or transport use after chemicals are satisfied.

Second, the -95% GHG constraint triggers a qualitative shift in S1/BES and S2/NFS. In the unconstrained cases the residual wood after chemicals is directed almost entirely to industrial process-heat boilers (25.9 TWh in S1, 16.8 TWh in S2, and 9.8 TWh in S3). Under the climate target, approximately 20.1 TWh of the available budget in S1 and S2 is instead converted to road diesel and aviation jet fuel – the hardest-to-electrify end uses – while industrial-heat allocation remains broadly stable. In S3/BDS, the 40 TWh ceiling is already fully committed to chemicals and industry heat under the climate constraint, leaving no margin for transport biofuels; the transport sector is instead partially decarbonised through imported renewable diesel, substituting the domestic biomass-to-fuel conversion route with a higher-cost import pathway.

Third, removing nuclear in S1/BES and S2/NFS amplifies biomass pressure across multiple services simultaneously. Beyond transport biofuels (which remain near 20 TWh), the model deploys an additional 15.6 TWh (S1) and 12.4 TWh (S2) through decentralised building-heat boilers, an additional 10.1 TWh (S1) and 8.8 TWh (S2) through industrial-heat boilers, and a new 8.3–8.6 TWh allocation to synthetic methane for seasonal storage and industrial gas demand. The result is that, in the absence of nuclear, biomass becomes a multi-service strategic resource simultaneously filling gaps in building heat, industrial heat, gas supply, and transport fuels. S3/BDS cannot follow this pattern: its end-use breakdown is structurally identical between the -95% and the no-nuclear case because the 40 TWh domestic ceiling is already binding under the weaker constraint. The additional decarbonisation burden – decentralised heat, synthetic gas – must be absorbed entirely by non-biomass pathways, which is why the cost penalty for nuclear removal is nearly three times larger in S3 than in S1.

The policy implication is therefore not only about the total quantity of forest biomass avail-

able, but about which end-use priorities the forest budget can still accommodate once both biodiversity constraints and nuclear policy choices tighten simultaneously. Figure 11 adds the end-use interpretation behind that aggregate picture. In S1/BES and S2/NFS, the no-nuclear variants redirect more biomass toward the hardest-to-abate services, indicating that biomass is treated as a strategic balancing resource once zero-carbon baseload flexibility declines. By contrast, S3/BDS cannot respond to the same pressure by expanding forest harvest, so the model is forced to reallocate scarce biomass more selectively and lean more heavily on non-biomass substitutions elsewhere in the system. The policy implication is that the debate is not only about the total quantity of forest biomass available, but also about which end uses receive priority when both climate ambition and forest constraints tighten.

4.3 Next steps: disturbance shocks and dynamic biomass supply

The present paper deliberately stops at static 2035 management scenarios. The next stage of the research is to couple the Finnish forest simulator used by the Jyväskylä/LUKE team with the FLAM and PICUS disturbance modules developed in the ForestNavigator framework, so that annual biomass supply curves can respond to wildfire risk, bark-beetle outbreaks, windthrow, salvageable calamity wood, and the subsequent reduction in standing stock (Venäläinen et al., 2020; Hlásny et al., 2021; Toth et al., 2020). In energy-system terms, this would extend the current forest scenarios from fixed 2035 ladders to time-varying biomass availability and cost trajectories under contrasting management regimes and climate pathways.

The concept-note design is not simply a lower biomass ceiling. It is intended to capture three mechanisms that are absent from the current paper: a short-lived calamity-wood pulse after a disturbance event, quality and marginal-cost changes for salvage biomass, and a medium-term decline in future harvestable stock after damaged cohorts are depleted (Toth et al., 2020; Rečka et al., 2023). Methodologically, the proposed extension uses a pre-shock 2030 EnergyScope snapshot, followed by a post-shock snapshot with limited re-optimisation in order to reflect realistic investment lead times for biomass CHP, DHN extensions, and other inherited infrastructure. The disturbance module is therefore presented as the logical next research phase built on the present article: the current paper establishes the calibrated national baseline and the static forest-scenario framework, while the coupled forest-disturbance workflow would add dynamic timing, quality, and price effects on top of that validated base.

5 Conclusion

This paper develops a national EnergyScope TD application for Finland centred on the role of forest biomass in a 2035 transition setting. By translating contested forest-management narratives into explicit biomass supply curves, it links the Finnish forest-policy debate to whole-energy-system optimisation. The calibrated 2017 benchmark shows that the model can reproduce the main Finnish primary-energy aggregates, electricity balance, and net CO₂ emissions with high fidelity, while keeping the remaining structural mismatches transparent, especially for district heating and gas-related CHP co-production. This validation step is essential because it establishes that the Finnish implementation is sufficiently credible for scenario exploration before moving to forward-looking policy questions.

For the 2035 scenario matrix, the central result is not that biodiversity-constrained forest management makes deep decarbonisation impossible, but that it removes forest biomass as a low-cost adjustment margin exactly when climate ambition becomes stringent. The unconstrained 2035 cases are already strongly decarbonised, whereas the -95% cases reveal much sharper competition for biomass, particularly when nuclear is removed. S1/BES and S2/NFS retain some flexibility to expand forest-wood use, but S3/BDS saturates its conservative 40 TWh ceiling and becomes the highest-cost configuration. The paper therefore shows that forest-biomass assumptions are not a secondary sensitivity: they materially shape the cost and allocation logic of deep decarbonisation in Finland. The next research step is to extend this validated static

framework toward disturbance-driven annual biomass supply trajectories, so that wind, beetle, and wildfire shocks can be studied within the same energy-system optimisation framework.

References

- W.R.L. Anderegg, C. Wu, N. Acil, N. Carvalhais, T.A.M. Pugh, J.P. Sadler, and R. Seidl. A climate risk analysis of Earth’s forests in the 21st century. *Science*, 377(6610), 2022. doi: 10.1126/science.abp9723.
- Tim Beringer, Wolfgang Lucht, and Sibyll Schaphoff. Bioenergy production potential of global biomass plantations under environmental and agricultural constraints. *GCB Bioenergy*, 3(4):299–312, 2011. ISSN 1757-1707. doi: 10.1111/j.1757-1707.2010.01088.x. URL <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1757-1707.2010.01088.x>. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1757-1707.2010.01088.x>.
- Clemens Blattert, Kyle Eyvindson, Markus Hartikainen, Daniel Burgas, Maria Potterf, Jani Lukkarinen, Tord Snäll, Astor Toraño-Caicoya, and Mikko Mönkkönen. Sectoral policies cause incoherence in forest management and ecosystem service provisioning. *Forest Policy and Economics*, 136:102689, March 2022. ISSN 1389-9341. doi: 10.1016/j.forpol.2022.102689. URL <https://www.sciencedirect.com/science/article/pii/S1389934122000016>.
- M. Borasio and S. Moret. Deep decarbonisation of regional energy systems: A novel modelling approach and its application to the Italian energy transition. *Renewable and Sustainable Energy Reviews*, 153:111730, January 2022. ISSN 13640321. doi: 10.1016/j.rser.2021.111730. URL <https://linkinghub.elsevier.com/retrieve/pii/S1364032121010030>.
- Víctor Codina Gironès, Emanuela Peduzzi, François Vuille, and François Maréchal. On the Assessment of the CO2 Mitigation Potential of Woody Biomass. *Frontiers in Energy Research*, 5, January 2018. ISSN 2296-598X. doi: 10.3389/fenrg.2017.00037. URL <https://www.frontiersin.org/journals/energy-research/articles/10.3389/fenrg.2017.00037/full>.
- Martin Colla, Julien Blondeau, and Hervé Jeanmart. Optimal Use of Lignocellulosic Biomass for the Energy Transition, Including the Non-Energy Demand: The Case of the Belgian Energy System. *Frontiers in Energy Research*, 10, March 2022. ISSN 2296-598X. doi: 10.3389/fenrg.2022.802327. URL <https://www.frontiersin.org/journals/energy-research/articles/10.3389/fenrg.2022.802327/full>.
- Martin Colla, Kevin Verleysen, Julien Blondeau, and Hervé Jeanmart. Navigating bioenergy horizons: a critical examination of Europe’s potential, with Belgium as a case study. *Sustainable Energy Research*, 11(1):17, May 2024. ISSN 2731-9237. doi: 10.1186/s40807-024-00108-0. URL <https://doi.org/10.1186/s40807-024-00108-0>.
- Vassilis Daioglou, Birka Wicke, Andre P. C. Faaij, and Detlef P. van Vuuren. Competing uses of biomass for energy and chemicals: implications for long-term global CO2 mitigation potential. *GCB Bioenergy*, 7(6):1321–1334, 2015. ISSN 1757-1707. doi: 10.1111/gcbb.12228. URL <https://onlinelibrary.wiley.com/doi/abs/10.1111/gcbb.12228>. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/gcbb.12228>.
- Richard Fischer, Eliza Zhunusova, Sven Günter, Susanne Iost, Franziska Schier, Jörg Schweinle, Holger Weimar, and Matthias Dieter. Leakage of biodiversity risks under the European Union Biodiversity Strategy 2030. *Conservation Biology*, 38(3):e14235, June 2024. ISSN 0888-8892, 1523-1739. doi: 10.1111/cobi.14235. URL <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.14235>.
- Helmut Haberl, Tim Beringer, Sribas C Bhattacharya, Karl-Heinz Erb, and Monique Hoogwijk. The global technical potential of bio-energy in 2050 considering sustainability constraints.

- Current Opinion in Environmental Sustainability*, 2(5):394–403, December 2010. ISSN 1877-3435. doi: 10.1016/j.cosust.2010.10.007. URL <https://www.sciencedirect.com/science/article/pii/S1877343510001132>.
- Aurélia Hernandez, Paolo Thiran, Hervé Jeanmart, Gauthier Limpens, Emmanuel DE Jaeger, and Julien Hendrickx. EnergyScope Multi-Cell : A novel open-source model for multi-regional energy systems. 2021.
- T. Hlásny, S. Zimová, K. Merganičová, P. Štěpánek, R. Modlinger, and M. Turčáni. Devastating outbreak of bark beetles in the Czech Republic: Drivers, impacts, and management implications. *Forest Ecology and Management*, 490:119075, June 2021. ISSN 0378-1127. doi: 10.1016/j.foreco.2021.119075. URL <https://www.sciencedirect.com/science/article/pii/S037811272100164X>.
- IEA. Bioenergy Annual Report 2023. Technical report, IEA Bioenergy, 2023. URL <https://www.ieabioenergy.com/wp-content/uploads/2024/06/Annual-Report-2023.pdf>.
- International Energy Agency IEA. Key World Energy Statistics 2020. page 81, 2020.
- A. Maarit I. Kallio and Meley M. Rannestad. Potential impacts of the EU’s biodiversity strategy on the EU and global forest sector and on the risk of biodiversity loss. *Environment Systems and Decisions*, 45(4):58, September 2025. ISSN 2194-5411. doi: 10.1007/s10669-025-10050-1. URL <https://doi.org/10.1007/s10669-025-10050-1>.
- Judit Lecina-Diaz, Jordi Martínez-Vilalta, Albert Alvarez, Mireia Banqué, Jörn Birkmann, Daniel Feldmeyer, Jordi Vayreda, and Javier Retana. Characterizing forest vulnerability and risk to climate-change hazards. *Frontiers in Ecology and the Environment*, 19(2): 126–133, March 2021. ISSN 1540-9295, 1540-9309. doi: 10.1002/fee.2278. URL <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/fee.2278>.
- Gauthier Limpens. Generating energy transition pathways: application to Belgium. 2021. doi: 10.13140/RG.2.2.25755.18724. URL <https://www.researchgate.net/doi/10.13140/RG.2.2.25755.18724>.
- Gauthier Limpens, Stefano Moret, Hervé Jeanmart, and Francois Maréchal. EnergyScope TD: A novel open-source model for regional energy systems. *Applied Energy*, 255:113729, December 2019. ISSN 0306-2619. doi: 10.1016/j.apenergy.2019.113729. URL <https://www.sciencedirect.com/science/article/pii/S0306261919314163>.
- Gauthier Limpens, Xavier Rixhon, Francesco Contino, and Hervé Jeanmart. EnergyScope Pathway: An open-source model to optimise the energy transition pathways of a regional whole-energy system. *Applied Energy*, 358:122501, March 2024. ISSN 03062619. doi: 10.1016/j.apenergy.2023.122501. URL <https://linkinghub.elsevier.com/retrieve/pii/S0306261923018652>.
- Stefano Moret. Strategic energy planning under uncertainty. *École Polytechnique Fédérale de Lausanne*, 2017. doi: 10.5075/EPFL-THESIS-7961.
- Stefano Moret, Víctor Codina Gironès, Michel Bierlaire, and François Maréchal. Characterization of input uncertainties in strategic energy planning models. *Applied Energy*, 202: 597–617, September 2017. ISSN 0306-2619. doi: 10.1016/j.apenergy.2017.05.106. URL <https://www.sciencedirect.com/science/article/pii/S0306261917306116>.
- Mikko Mönkkönen, Clemens Blattert, Jérémy Cours, Rémi Duflot, Merja Elo, Kyle Eyvindson, Jari Kouki, María Triviño, and Daniel Burgas. Ecologically and economically sustainable level of timber harvesting in boreal forests – defining the safe operating space for forest use, July 2024. URL <http://biorxiv.org/lookup/doi/10.1101/2024.06.27.600997>.

Mikko Mönkkönen, Clemens Blattert, Jérémy Cours, Rémi Duflot, Merja Elo, Kyle Eyvindson, Jari Kouki, María Triviño, and Daniel Burgas. Ecologically and economically sustainable level of timber harvesting in boreal forests – defining the safe operating space for forest use, January 2025. URL <https://www.biorxiv.org/content/10.1101/2024.06.27.600997v2>. Pages: 2024.06.27.600997 Section: New Results.

Natural Resources Institute Finland (Luke). Wood in energy generation 2024, 2024.

Natural Resources Institute Finland (Luke). Foreign trade in roundwood and forest industry products by country 2024 (provisional), April 2025. URL <https://www.luke.fi/en/statistics/foreign-trade-in-roundwood-and-forest-industry-products-statistics-discontinued/foreign-trade-in-roundwood-and-forest-industry-products-by-country-2024-provisional>.

Lukáš Rečka, Milan Ščasný, and Dali Tsintsikladze Laxton. The Role of Biomass in Decarbonisation Efforts: Spatially Enriched Energy System Optimisation Modelling. *Energies*, 16(21):7380, October 2023. ISSN 1996-1073. doi: 10.3390/en16217380. URL <https://www.mdpi.com/1996-1073/16/21/7380>.

Francesca Rosa, Fulvio Di Fulvio, Pekka Lauri, Adam Felton, Nicklas Forsell, Stephan Pfister, and Stefanie Hellweg. Can Forest Management Practices Counteract Species Loss Arising from Increasing European Demand for Forest Biomass under Climate Mitigation Scenarios? *Environmental Science & Technology*, 57(5):2149–2161, February 2023. ISSN 0013-936X, 1520-5851. doi: 10.1021/acs.est.2c07867. URL <https://pubs.acs.org/doi/10.1021/acs.est.2c07867>.

P. Ruiz, W. Nijs, D. Tarvydas, A. Sgobbi, A. Zucker, R. Pilli, R. Jonsson, A. Camia, C. Thiel, C. Hoyer-Klick, F. Dalla Longa, T. Kober, J. Badger, P. Volker, B.S. Elbersen, A. Brosowski, and D. Thrän. ENSPRESO - an open, EU-28 wide, transparent and coherent database of wind, solar and biomass energy potentials. *Energy Strategy Reviews*, 26:100379, November 2019. ISSN 2211467X. doi: 10.1016/j.esr.2019.100379. URL <https://linkinghub.elsevier.com/retrieve/pii/S2211467X19300720>.

Franziska Schier, Susanne Iost, Björn Seintsch, Holger Weimar, and Matthias Dieter. Assessment of Possible Production Leakage from Implementing the EU Biodiversity Strategy on Forest Product Markets. *Forests*, 13(8), August 2022. ISSN 1999-4907. doi: 10.3390/f13081225. URL <https://www.mdpi.com/1999-4907/13/8/1225>.

Vivian Schueler, Ulf Weddige, Tim Beringer, Liliana Gamba, and Patrick Lamers. Global biomass potentials under sustainability restrictions defined by the European Renewable Energy Directive 2009/28/EC. *GCB Bioenergy*, 5(6):652–663, 2013. ISSN 1757-1707. doi: 10.1111/gcbb.12036. URL <https://onlinelibrary.wiley.com/doi/abs/10.1111/gcbb.12036>. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/gcbb.12036>.

Stephanie Searle and Chris Malins. A reassessment of global bioenergy potential in 2050. *GCB Bioenergy*, 7(2):328–336, 2015. ISSN 1757-1707. doi: 10.1111/gcbb.12141. URL <https://onlinelibrary.wiley.com/doi/abs/10.1111/gcbb.12141>. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/gcbb.12141>.

Bernhard Steubing, Rainer Zah, and Christian Ludwig. Heat, Electricity, or Transportation? The Optimal Use of Residual and Waste Biomass in Europe from an Environmental Perspective. *Environmental Science & Technology*, 46(1):164–171, January 2012. ISSN 0013-936X. doi: 10.1021/es202154k. URL <https://doi.org/10.1021/es202154k>.

Paolo Thiran. *Exploring options for a fossil-free European energy system : The role of renewable fuels*. PhD thesis, UCL, 2025. URL <http://hdl.handle.net/2078.1/293132>.

- Daniel Toth, Mansoor Maitah, Kamil Maitah, and Veronika Jarolínová. The Impacts of Calamity Logging on the Development of Spruce Wood Prices in Czech Forestry. *Forests*, 11(3): 283, March 2020. ISSN 1999-4907. doi: 10.3390/f11030283. URL <https://www.mdpi.com/1999-4907/11/3/283>. Number: 3.
- Ari Venäläinen, Ilari Lehtonen, Mikko Laapas, Kimmo Ruosteenoja, Olli-Pekka Tikkanen, Heli Viiri, Veli-Pekka Ikonen, and Heli Peltola. Climate change induces multiple risks to boreal forests and forestry in Finland: A literature review. *Global Change Biology*, 26(8):4178–4196, August 2020. ISSN 1354-1013, 1365-2486. doi: 10.1111/gcb.15183. URL <https://onlinelibrary.wiley.com/doi/10.1111/gcb.15183>.
- Bertil Wahlund, Jinyue Yan, and Mats Westermark. Increasing biomass utilisation in energy systems : a comparative study of CO2 reduction and cost for different bioenergy processing options. *Biomass and Bioenergy*, 26(6):531–544, 2004. doi: 10.1016/j.biombioe.2003.09.003. URL <https://urn.kb.se/resolve?urn=urn:nbn:se:ltu:diva-10672>.
- Omid Zamani and Janine Pelikan. Deforestation-free trade: the impact of border regulations on trade, production and land use in the forestry and agricultural sectors. *Environmental Research Letters*, 21(2):024009, January 2026. ISSN 1748-9326. doi: 10.1088/1748-9326/ae2f71. URL <https://doi.org/10.1088/1748-9326/ae2f71>.
- Václav Šimůnek, Zdeněk Vacek, Stanislav Vacek, Michal Švanda, Vojtěch Hájek, and Giuseppe D’Andrea. Norway spruce forest management in the Czech Republic is linked to the solar cycle under conditions of climate change – from tree rings to salvage harvesting. *Journal of Space Weather and Space Climate*, 14:37, 2024. ISSN 2115-7251. doi: 10.1051/swsc/2024030. URL <https://www.swsc-journal.org/articles/swsc/abs/2024/01/swsc230016/swsc230016.html>.